Artificial Intelligence Literacy and Readiness in Future Health Care Professionals: A Cross-sectional Study

Abstract

To examine the relationship between artificial intelligence (AI) literacy and readiness for medical AI among medical and nursing students, and to explore demographic and educational factors affecting AI literacy and readiness for AI. This cross-sectional study enrolled 443 students attending the Faculty of Medicine and the Department of Nursing at Bilecik Seyh Edebali University between May and June 2025. Data were collected using a general demographic questionnaire, Medical Artificial Intelligence Readiness Scale (MAIRS), and Artificial Intelligence Literacy Scale (AILS). Medical students showed higher AI readiness ($P < 0.001$) and literacy ($P = 0.032$) scores than nursing students. AI literacy was moderately and positively correlated with readiness for medical AI ($r = 0.338$, $P < 0.001$), and the associations were strongest on the Awareness and Usage dimensions. In multivariable models, MAIRS scores were independently associated with the study year, department, and single-item readiness to use AI, while AILS scores were independently associated with the readiness to use AI and the belief that AI could partially replace health care workers.Internal consistency of the questionnaires was high (MAIRS = 0.972; AILS = 0.878). The findings support the integration of structured, practical, and ethically informed AI education into health science curricula, as this may be associated with

greater student readiness for future AI-supported health care environments.